FEASIBILITY ANALYSIS · RETRIEVAL-AUGMENTED CULTURAL QA

Can a CultureBank graph answer CulturalBench?

A pre-registration check run against both datasets in full: the topics line up almost perfectly, the countries do not, and the facts that decide each question are mostly absent. Here is what the numbers say before you spend a single inference call.

Sources CultureBank (22,990 descriptors) · CulturalBench-Easy (1,227 questions)

Method full-dataset lexical analysis, no model calls Date 4 Sep 2026

THE SHORT ANSWER

Topic coverage is not your problem. Country coverage and fact granularity are. Every topic CulturalBench asks about exists in CultureBank in volume — food, greetings, festivals, family, money, dress. On topics alone the two datasets look like a match.

But CultureBank's mass sits where CulturalBench barely asks. The United States is **31% of CultureBank and 1.6% of CulturalBench**. Bangladesh, Nepal, Zimbabwe, Hong Kong and Iran carry 136 questions between them and are supported by **67 descriptors in total**.

The decisive test: even with `oracle` retrieval — searching each country's descriptors using the correct answer itself — CultureBank picks the gold option over the three distractors only **27.9%** of the time against a 25% chance baseline. The discriminating facts are largely not in there.

That said, the signal is real and strongly graded by density. Where CultureBank is rich, oracle accuracy reaches **34–49%**; where it is starved it falls to **14.6%**, below chance, meaning retrieval there actively pulls toward wrong answers. **A country-gated graph is worth building. An always-on one will cost you accuracy.**

01 Topics: a clean match

Classifying all 1,227 questions and all 22,990 descriptors into a shared 15-topic scheme, no topic CulturalBench asks about is missing from CultureBank.

CultureBank ships its own 36-topic vocabulary; CulturalBench-Easy ships no topic labels at all. To compare them I classified both sides independently with a keyword lexicon, so the mapping is not circular. Every bucket has supply, and most have supply within 2× of demand.

Topic supply and demand

Share of CulturalBench questions vs. share of CultureBank descriptors, by topic

TOPIC	QUESTIONS	DESCRIPTORS	SUPPLY / DEMAND
Food & dining	28.9%	13.8%	**0.48×**
Greeting & etiquette	14.1%	7.1%	**0.50×**
Celebrations & festivals	11.2%	8.5%	**0.76×**
Family & relationships	8.6%	11.1%	1.29×
Gift giving & hospitality	5.8%	4.6%	**0.79×**
Language & communication	3.8%	10.4%	2.74×
Work & workplace	3.7%	4.7%	1.27×
Dress & appearance	3.1%	3.3%	1.06×
Transport & public space	3.1%	3.3%	1.06×
Money, tipping & shopping	2.7%	5.1%	1.89×
Religion & beliefs	2.6%	2.8%	1.08×
Education & school	1.7%	5.3%	3.12×
Home & daily life	1.6%	2.6%	1.63×
Health & body	1.1%	3.9%	3.55×
Death & taboo	0.3%	0.4%	1.33×

Ratios below 1.0 (rust) mean CulturalBench asks about the topic more than CultureBank covers it. Food and greetings are the two under-supplied areas — and together they are 43% of the benchmark. Nothing is at zero.

The two under-supplied buckets matter because they are the benchmark's centre of gravity. Food and dining alone is **28.9% of all questions** but 13.8% of descriptors; greeting and etiquette is 14.1% against 7.1%. Conversely CultureBank over-supplies language, education and health — topics CulturalBench largely ignores. So roughly a quarter of your graph is built on subject matter the benchmark will never ask about.

02 Countries: an inverse relationship

The countries CulturalBench asks about most are the ones CultureBank knows least, almost monotonically.

CultureBank's 2,410 free-text cultural groups map onto CulturalBench's 45 countries for 74.9% of descriptors (the remainder are groups like `European`, `Swedish` or `Muslims` that fall outside the benchmark's country list). Mapped, the distribution is stark.

31.4%
of CultureBank is United States — which is 1.6% of CulturalBench

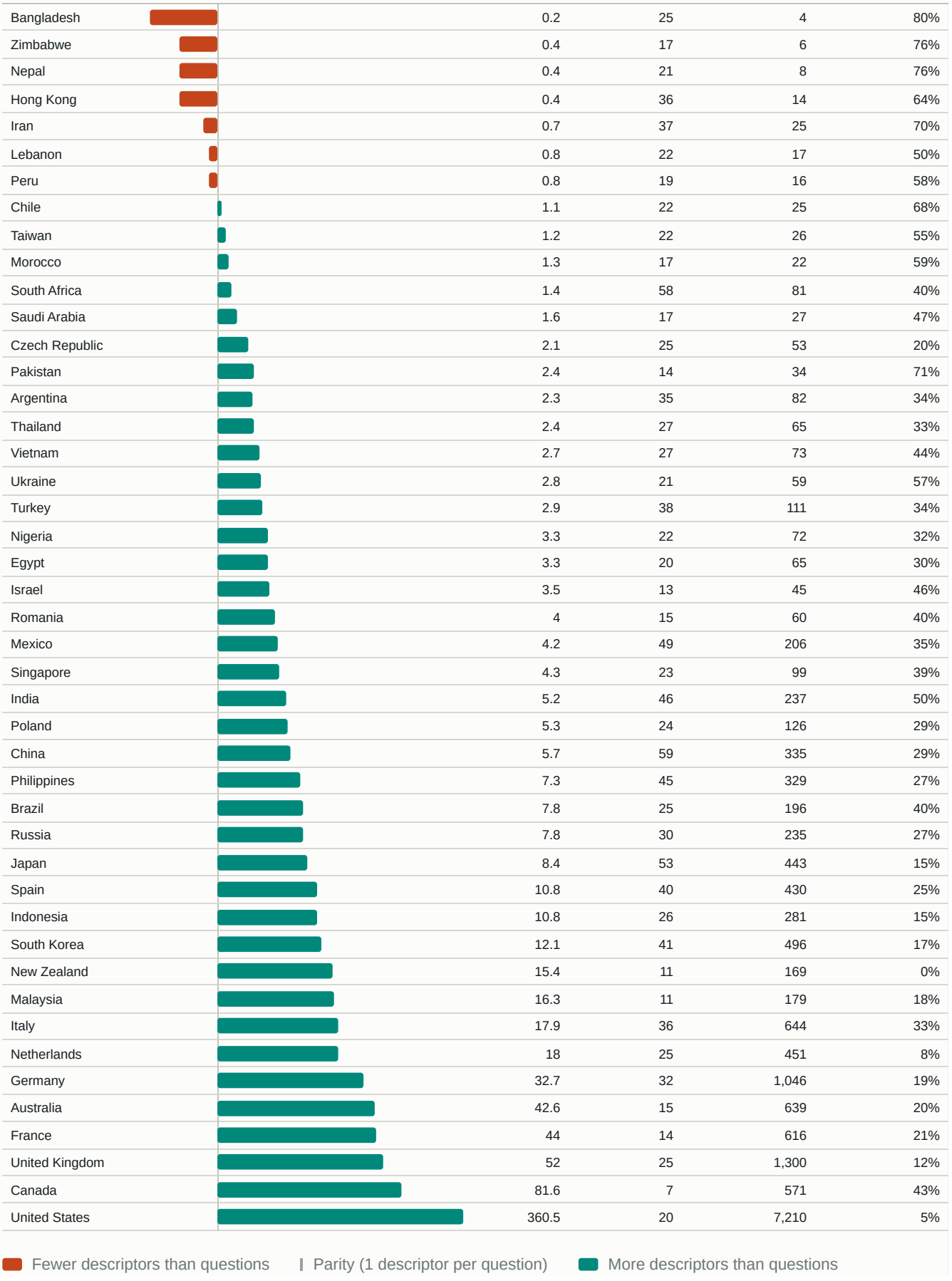
23%
of questions target countries with fewer than 50 descriptors

61.6%
of questions fall in a country × topic cell with under 20 descriptors

6.7%
fall in a cell with literally zero descriptors

Coverage gazetteer

Descriptors available per benchmark question, log scale, all 45 countries



COUNTRY	DESC/Q	QUESTIONS	DESCRIPTORS	GOLD ABSENT
Bangladesh	0.2	25	4	80%
Zimbabwe	0.4	17	6	76%
Nepal	0.4	21	8	76%
Hong Kong	0.4	36	14	64%
Iran	0.7	37	25	70%
Lebanon	0.8	22	17	50%
Peru	0.8	19	16	58%
Chile	1.1	22	25	68%
Taiwan	1.2	22	26	55%
Morocco	1.3	17	22	59%
South Africa	1.4	58	81	40%
Saudi Arabia	1.6	17	27	47%
Czech Republic	2.1	25	53	20%
Pakistan	2.4	14	34	71%
Argentina	2.3	35	82	34%
Thailand	2.4	27	65	33%
Vietnam	2.7	27	73	44%
Ukraine	2.8	21	59	57%
Turkey	2.9	38	111	34%
Nigeria	3.3	22	72	32%
Egypt	3.3	20	65	30%
Israel	3.5	13	45	46%
Romania	4	15	60	40%
Mexico	4.2	49	206	35%
Singapore	4.3	23	99	39%
India	5.2	46	237	50%
Poland	5.3	24	126	29%
China	5.7	59	335	29%
Philippines	7.3	45	329	27%
Brazil	7.8	25	196	40%
Russia	7.8	30	235	27%
Japan	8.4	53	443	15%
Spain	10.8	40	430	25%
Indonesia	10.8	26	281	15%
South Korea	12.1	41	496	17%
New Zealand	15.4	11	169	0%
Malaysia	16.3	11	179	18%
Italy	17.9	36	644	33%
Netherlands	18	25	451	8%
Germany	32.7	32	1,046	19%
Australia	42.6	15	639	20%
France	44	14	616	21%
United Kingdom	52	25	1,300	12%
Canada	81.6	7	571	43%
United States	360.5	20	7,210	5%

Fewer descriptors than questions | Parity (1 descriptor per question) | More descriptors than questions

Desc/Q is CultureBank descriptors divided by benchmark questions for that country; the bar is its log. **Gold absent** is the share of that country's questions where none of the correct answer's content words appear anywhere in that country's descriptors. The bottom twelve rows are where a graph would have almost nothing to say.

Read the top of that table as the risk register. **Bangladesh: 25 questions, 4 descriptors.** Zimbabwe: 17 questions, 6 descriptors. Hong Kong: 36 questions, 14 descriptors. I checked these by hand against every spelling variant in the data — `Bangladeshi`, `Bengali`, `Dhaka`, `Shona`, `Cantonese`, `Hong Kongers` — and the counts hold. These countries are genuinely absent, not mis-mapped.

The cell-level view is harsher than the country view. `Bangladesh × Food` is 8 questions against 0 descriptors. `Iran × Food` is 11 against 1. `Hong Kong × Food` is 12 against 3. Meanwhile `United Kingdom × Food` has 264 descriptors serving 9 questions.

03 The oracle test

Coverage counts tell you something is there. They do not tell you it is the right thing. This is the test that separates the two.

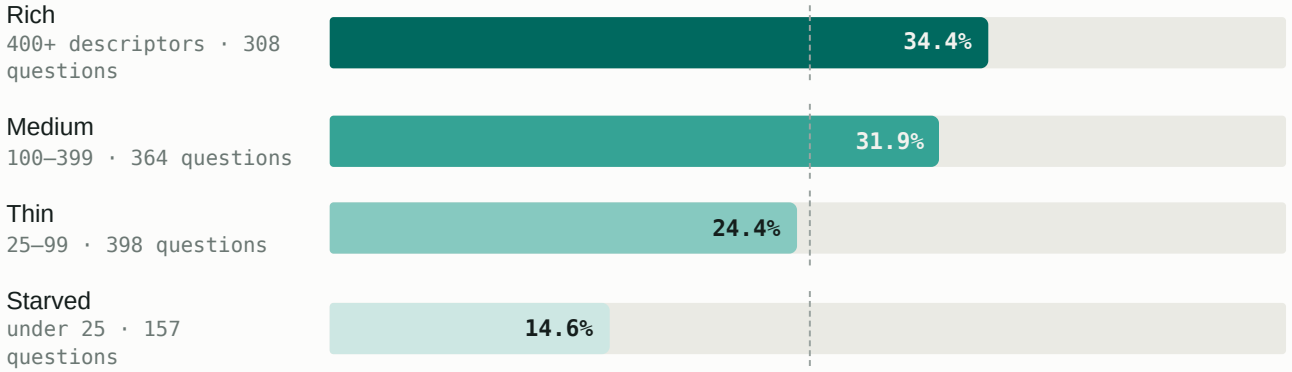
A retrieval system can fail two ways: the retriever misses a fact that is present, or the fact was never present. To isolate the second, I ran an oracle: for each question, score all four options against that country's entire descriptor set, using the option text itself as the query. This is retrieval that cannot fail — it already knows what it is looking for. If CultureBank contains the discriminating fact, the gold option should out-score the distractors.

27.9%	11.7%	36.9%	77.9%
oracle picks the gold option (chance = 25%)	of gold answers are fully present in some descriptor	of gold answers are entirely absent from their country's data	of questions have a distractor at least as findable as the gold

27.9% against a 25% floor is close to nothing. But the aggregate hides a clean gradient — the signal is entirely a function of how much CultureBank knows about that country.

Oracle accuracy by CultureBank density

Share of questions where CultureBank's own text favours the correct answer



Dashed line: 25% random baseline (four options)

Countries grouped by descriptor count. The gradient is monotonic and spans 20 points. The starved tier sits **10 points below chance**: in 55 of those 157 questions a distractor is strictly more findable than the gold answer, so retrieval there does not merely fail to help — it argues for the wrong option.

The signal is real, not an artefact

Before trusting any of this I ran the obvious control. Using BM25 over the *correct* country's descriptors, the retrieved text favours the gold option 36–38% of the time. Re-run identically but retrieving from a *randomly chosen wrong* country, that drops to **25.6%** — exactly chance. The lift is genuinely country-specific cultural content, not an artefact of option length or word frequency. CultureBank does know things. It mostly does not know the things this benchmark tests.

04 Why: a granularity mismatch

CultureBank descriptors are aggregated behavioural generalisations. CulturalBench questions turn on specific discriminating facts. These are different objects.

Reading the actual retrievals makes the failure mode obvious in a way the statistics do not. Here are three real top-1 results, country-matched, from the questions where retrieval scored highest.

SPAIN · FOOD & DINING

Which is the preferred way of cooking eggs for Spanish people?

NEEDS Omelette — a specific dish fact

RETRIEVED "In the Spanish cultural context, especially during cooking activities with family members, particularly mothers and daughters, there is a strong emphasis on engaging …"

MOROCCO · GREETING & ETIQUETTE

In Moroccan culture, how do people typically greet people they know on the street?

NEEDS The phrase "peace be upon you"

RETRIEVED "In Morocco and Moroccan-related events, Moroccan people and society engage in music-related activities, including listening to and creating music …"

UNITED STATES · SOCIAL NORMS

If there is only one slice of pizza left in the last box, what is the most common action someone would take?

NEEDS Offer it to someone else first

RETRIEVED "In the United States … the American cultural group use honey not only as a sweetener and condiment but also in unconventional ways, such as on pizza"

That last one is scored as a *hit* by the automatic metric — the word "pizza" matched. It is the clearest illustration of why the 36% BM25 number is optimistic and the 27.9% oracle number is the one to trust. CultureBank's descriptors are the right *topic* at the wrong *resolution*: they say Spaniards cook with family, where the question needs *tortilla española*. That resolution gap is structural, and it comes straight from CultureBank's construction — it was built by clustering and summarising community comments about culture shock, which surfaces broad generalisations, not the specific conventions a native would name.

05 What I would do with this

The finding does not say abandon the experiment. It says redesign it so the result is interpretable either way.

1 **Gate retrieval on country density, and measure the gate**

Route to the graph only when the country has meaningful support — roughly 100+ descriptors, which covers 672 of 1,227 questions — and fall back to the base model otherwise. Report gated and ungated separately. Ungated, you should expect the starved tier to *drag your score below baseline*, and a single averaged number will hide that.

2

Evaluate on Hard, not Easy

CulturalBench-Easy is close to saturated — the best reported model sits at 89.6% against a 92.4% human baseline, so there is under three points of headroom and no room to detect a real effect. CulturalBench-Hard, where frontier models sit near 61%, is where a knowledge intervention can actually move a number.

3

Check the option-order confound first

In the released CulturalBench-Easy file, answer A appears in 658 of 1,227 rows — **53.6%**. If your harness does not shuffle options, "always answer A" beats everything measured in this report, and any gain you observe may be a positional artefact. Shuffle, and fix the seed.

4

Run a dense-retrieval replication before concluding

Everything here is lexical, which is a floor, not a ceiling. Embeddings would recover paraphrase — "don't tip" entailing "tipping generously" is false — that BM25 misses entirely. Re-run the oracle test with an embedding model; if it stays near 25%, the knowledge genuinely is not there and the lexical result is confirmed.

5

Consider whether CulturalBench is the right target at all

CultureBank's own paper evaluates on GlobalOpinionQA and CultureNLI, and reports gains of roughly 2 points. Those benchmarks test aggregate cultural tendencies, which is the shape of knowledge CultureBank actually holds. CulturalBench tests specific conventions. A negative result here may say more about the pairing than about your graph.

6

Use the graph for the thing a flat corpus can't do

If your KG only does country-filtered passage retrieval, it is a search index with extra steps, and this report predicts its ceiling. The structure earns its keep on multi-hop and contrast queries — bridging related groups to cover a starved country from its neighbours, or using the `agreement` and `num_support` fields to weight consensus. Design the ablation so it isolates that.

What this analysis cannot tell you

- **Lexical absence is not semantic absence.** A descriptor stating "the Dutch rarely tip" fully answers a question whose gold option is "tipping generously," and my method scores it as a miss. The 36.9% absence figure is therefore an upper bound on true absence.
- **Lexical presence is not support.** The reverse error is equally live — a matched word may sit inside a descriptor that asserts the opposite. Both errors push the oracle number around; only a model-in-the-loop check settles it.
- **The topic classifier is mine, not either dataset's.** CulturalBench-Easy ships no topic labels, so the 15-bucket scheme is a keyword lexicon I wrote. It left 7.8% of questions unclassified and will have misfiled some others.
- **Group-to-country mapping is lossy.** 25.1% of descriptors map to no benchmark country. I folded sub-national groups upward (Californians → US, Cantonese → Hong Kong), which is a judgement call that inflates the large countries slightly.
- **The released CultureBank TikTok split has 11,754 rows**, not the 12,754 the paper reports. I used the released file.

Analysis run 4 September 2026 over the complete released files: SALT-NLP/CultureBank (TikTok + Reddit) and kellycyy/CulturalBench (Easy split). Papers: Shi et al., EMNLP Findings 2024 · Chiu et al., ACL 2025. No model inference was used; all figures are reproducible from the two files with BM25 and a keyword lexicon.